Mughesh Kumar N R

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Education

University of Virginia *Aug 2025 – Dec 2026*
Master of Science in Computer Science Charlottesville, VA
- Completed graduate coursework in Data Structures, Algorithms, Machine Learning, and Cyber-Physical Systems.
- Served as a Graduate Teaching and Research Assistant, guiding student projects and supporting professor's research.
- Judged student hackathons and led campus ethics initiatives, organizing campus events to promote ethical technology.

University of California, Berkeley *Jan 2025 – May 2025*
Startup Semester Program Berkeley, CA
- Studied Designing Startups to Transform Society, Future of Technology, Product Management, and AI in Business.
- Awarded **$10,950** UC Berkeley Chancellor's Scholarship for Startup Semester, Selected from 600+ applicants.

SRM Institute of Science and Technology *Aug 2021 – Dec 2024*
Bachelor of Technology in Computer Science and Engineering (AI & ML) Chennai, India
- Awarded University Gold Medal for Leadership and Academic Excellence, and also served as Student Body President.
- Won two Sustainable Development Goals hackathons and $1,500 in awards, solving real-world problems in 48-hours.
- Published two IEEE research papers and received an Indian-granted patent in artificial intelligence and innovation.

Work Experience

University of Virginia Information Technology Services (ITS) *June 2026 – Aug 2026*
Wireless Systems Deployment Summer Intern Charlottesville, VA
- Supported a **$6M+ Wireless Network Refresh** project with **HPE** to modernize university wireless infrastructure.
- Conducted campus-wide wireless surveys and documented the geographic locations of **6,656** wireless access points.
- Utilized **ArcGIS** to inventory, map, and validate wireless assets, improving deployment planning and asset visibility.
- Developed floorplans, maps, and deployment data within the **Juniper Mist** wireless network management platform.

IBM *June 2023 – July 2023*
Artificial Intelligence Intern India
- Built a Mental Fitness Tracker using **Pandas** and **NumPy**, performing analytics on merged global health datasets.
- Applied statistics and mathematics for predictive modeling of mental-health DALYs from disorder prevalence features.
- Implemented Linear Regression and Random Forest with scikit-learn. Evaluated models using MSE, RMSE, and R^2.
- Improved prediction accuracy to 99.4% R^2 with Random Forest model, outperforming Linear Regression (76.6% R^2).

Projects

FaceMark AI — Smart Face Recognition Attendance System [Live Demo]
Tech Stack: *HTML, CSS, JavaScript, face-api.js, MTCNN, WebRTC, Canvas API, Web Speech API, Vercel.*
- Built a browser-based face recognition attendance system using HTML, CSS, and JavaScript with face-api.js and the MTCNN model, detecting faces in real time and generating 128-D face descriptors for accurate identity matching.
- Implemented real-time webcam capture with WebRTC (getUserMedia) and Canvas API overlays, along with a guided 30-sample training flow, a 30-minute cooldown to block duplicate entries, and automated Web Speech voice greetings.
- Designed a privacy-first Web-App with localStorage persistence, date filters, data visualization and CSV export.

Intelligent Computer Vision System for Pest Detection in Agri Farms
Tech Stack: *Python, TensorFlow, OpenCV, NoSQL, Microservices.*
- Designed and deployed a Computer Vision system using TensorFlow and OpenCV to detect and classify pests on crops, enabling precision agriculture and promoting sustainable farming practices across multiple regional agri farms.
- Detected 8+ pest species with 96% accuracy on real-time inference, training deep learning models on 12,000+ annotated crop images collected from large-scale field trials across diverse environmental and lighting conditions.
- Integrated Python-based microservices architecture with NoSQL databases for scalable data ingestion, storage, and real-time prediction serving, reducing end-to-end inference latency and enabling seamless deployment on Jetson Nano.

Technical Skills

Areas of Interest: AI, ML, Software Engineering, LLMs, Data Science, System Design, Network Engineering.
Languages: Python, C/C++, HTML/CSS, JavaScript, SQL.
Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, Flask, React, Node.js, Hugging Face, FastAPI, Next.js.
Libraries & Data: NumPy, Pandas, OpenCV, Matplotlib, Seaborn, Plotly, SciPy, Jupyter, MySQL, MongoDB, Redis.
Tools & Cloud: Git, Docker, Kubernetes, Linux, CI/CD, REST APIs, Agile, AWS (EC2, S3, Lambda), GCP, Azure.

Publications & Patents

- Optimizing Energy Efficiency in Smart Street Lighting Systems with LoRa WAN Technology – IEEE. Publication
- Comparative Analysis of Handwritten Digit Recognition using MLP, CNN, LeNet-5 Models – IEEE. Publication
- Internet of Things (IoT) Based Attendance System for Roadside Construction Sites – IP India Granted. Patent